Week 2 complete solutions

Problem-sheet solutions

Each solution identifies the intended method, gives intermediate work, and shows the mark allocation.

`W02-Q01-PA` (M4)

Method and conditions. The inner dimensions agree: $(2 \times 3)(3 \times 2)$, so the product exists and is 2×2.

$$\begin{aligned}\mathbf{AB} &= \begin{bmatrix} 1(2)+2(-1)+(-1)(3) & 1(0)+2(1)+(-1)(2) \\ 0(2)+3(-1)+1(3) & 0(0)+3(1)+1(2) \end{bmatrix} \\ &= \begin{bmatrix} -3 & 0 \\ 0 & 5 \end{bmatrix}.\end{aligned}$$

Also,

$$\mathbf{Bu} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \qquad \mathbf{A}(\mathbf{Bu}) = \begin{bmatrix} -3 \\ -5 \end{bmatrix}.$$

Using the product already computed,

$$(\mathbf{AB})\mathbf{u} = \begin{bmatrix} -3 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -3 \\ -5 \end{bmatrix}.$$

Final result. The product is the 2×2 matrix shown above, and the equal vectors verify $\mathbf{A}(\mathbf{Bu}) = (\mathbf{AB})\mathbf{u}$ in this case.

Marks (5). Correct dimension, 1; correct matrix product, 2; correct $\mathbf{Bu}$ and left-associated product, 1; correct right-associated product and comparison, 1.

Common mistakes. Multiplying entries position by position; omitting the dimension check; losing a negative sign; comparing products with incompatible parentheses.

`W02-Q01-PB` (M4)

Method and conditions. The reversed product has dimensions $(3 \times 2)(2 \times 3)$, so it exists and is 3×3.

$$\mathbf{BA} = \begin{bmatrix} 2 & 4 & -2 \\ -1 & 1 & 2 \\ 3 & 12 & -1 \end{bmatrix}.$$

For example, the second row is $-1[1, 2, -1] + 1[0, 3, 1] = [-1, 1, 2]$.

Transposing this product gives

$$(\mathbf{BA})^{\mathsf{T}} = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 1 & 12 \\ -2 & 2 & -1 \end{bmatrix}.$$

Direct multiplication gives the same matrix:

$$\mathbf{A}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}} = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 1 & 12 \\ -2 & 2 & -1 \end{bmatrix}.$$

Final result and interpretation. The first product is 2×2, while the reversed product is 3×3. They therefore cannot be equal. This example demonstrates that matrix multiplication is not commutative.

Marks (6). Correct 3×3 dimension and product, 3; correct transpose verification, 2; valid non-commutativity explanation, 1.

Common mistakes. Reusing the first product; reporting a 2×2 answer; saying only that order matters without comparing the products.

`W02-Q02-PA` **(M4)**

Method and conditions. For a 2×2 matrix, compute the determinant and use the inverse formula only if the determinant is non-zero.

$$\det(\mathbf{C}) = 2(3) - 1(5) = 1.$$

Because the determinant is non-zero,

$$\mathbf{C}^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}.$$

Verification gives

$$\mathbf{C}\mathbf{C}^{-1} = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}_2.$$

Final result. $\det(\mathbf{C}) = 1$, and the inverse is the matrix above.

Marks (6). Determinant setup and value, 2; correct inverse, 2; correct verification, 2.

Common mistakes. Using $ad + bc$; changing the signs of diagonal rather than off-diagonal entries; applying the inverse formula without checking that the determinant is non-zero.

`W02-Q02-PB` **(M4)**

Method and conditions. Let the displayed right-hand-side matrix be $\mathbf{D}$. Since $\mathbf{C}$ is invertible,

$$\begin{aligned} \mathbf{X} &= \mathbf{C}^{-1}\mathbf{D} \\ &= \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 3 & 4 \end{bmatrix}. \end{aligned}$$

Verification gives

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}.$$

Final result.

$$\mathbf{X} = \begin{bmatrix} -1 & -1 \\ 3 & 4 \end{bmatrix}.$$

Because $\det(\mathbf{C}) \neq 0$, multiplication by $\mathbf{C}$ is invertible, so each right-hand side has a unique solution.

Marks (6). Correct inverse setup, 1; correct two solution columns, 2; correct matrix verification, 2; uniqueness explanation, 1.

Common mistakes. Multiplying on the wrong side; using $\mathbf{CD}$ instead of $\mathbf{C}^{-1}\mathbf{D}$; swapping the two solution columns; omitting the requested check.

`W02-Q03-PA` **(M4)**

Method and conditions. Row operations preserve the solution set. Start from the 4×5 augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 2 & -1 & 1 & 0 & -1 \\ 1 & 2 & -1 & 2 & 12 \\ 3 & 1 & 2 & -1 & 0 \end{array}\right].$$

Apply $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$ and $R_4 \leftarrow R_4 - 3R_1$, then interchange rows 2 and 3:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 0 & 1 & -2 & 1 & 7 \\ 0 & -3 & -1 & -2 & -11 \\ 0 & -2 & -1 & -4 & -15 \end{array}\right].$$

Use $R_3 \leftarrow R_3 + 3R_2$ and $R_4 \leftarrow R_4 + 2R_2$:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 0 & 1 & -2 & 1 & 7 \\ 0 & 0 & -7 & 1 & 10 \\ 0 & 0 & -5 & -2 & -1 \end{array}\right].$$

Finally use $R_4 \leftarrow 7R_4$, then $R_4 \leftarrow R_4 - 5R_3$:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 0 & 1 & -2 & 1 & 7 \\ 0 & 0 & -7 & 1 & 10 \\ 0 & 0 & 0 & -19 & -57 \end{array}\right].$$

Back-substitution gives $w = 3$, $z = -1$, $y = 2$ and $x = 1$.

Final result and classification.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}.$$

There are four pivots for four unknowns, so the system has a unique solution.

Substitution gives, in order, $1 + 2 - 1 + 3 = 5$, $2 - 2 - 1 = -1$, $1 + 4 + 1 + 6 = 12$, and $3 + 2 - 2 - 3 = 0$, verifying all four equations.

Marks (10). Correct augmented matrix, 1; valid first-column elimination, 2; valid later elimination to echelon form, 3; correct back-substitution, 2; classification, 1; verification, 1.

Common mistakes. Applying a row operation to the coefficients but not the right-hand side; sign errors after the row interchange; giving the solution without the requested classification or verification.

`W02-Q04-PA` **(M4)**

Method and conditions. Begin with

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 2 \\ 3 & 4 & 5 & k \end{array}\right].$$

Use $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & k-3 \end{array}\right].$$

Then $R_3 \leftarrow R_3 - R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & k-3 \end{array}\right].$$

The coefficient matrix has two pivots for every k, so $\operatorname{rank}(\mathbf{A}) = 2$. If $k = 3$, the last augmented row is zero and $\operatorname{rank}([\mathbf{A} \mid \mathbf{b}]) = 2$. If $k \neq 3$, the last augmented entry is a pivot and $\operatorname{rank}([\mathbf{A} \mid \mathbf{b}]) = 3$.

Final result.

	$\operatorname{rank}(\mathbf{A})$	$\operatorname{rank}([\mathbf{A} \mid \mathbf{b}])$
$k = 3$	2	2
$k \neq 3$	2	3

Marks (7). Correct augmented matrix and row-reduction route, 3; correct coefficient rank, 1; correct augmented ranks in both cases, 2; explicit case boundary, 1.

Common mistakes. Treating $k - 3$ as a coefficient pivot; reporting that the coefficient rank changes with k; failing to separate the two cases.

`W02-Q04-PB` **(M4)**

Method and conditions. A system is consistent exactly when the coefficient and augmented ranks agree.

If $k \neq 3$, the ranks are 2 and 3, so the system has no solution.

If $k = 3$, both ranks are 2. This is less than the number of unknowns, which is 3, so the system has multiple solutions. Let $z = t \in \mathbb{R}$. The echelon equations give

$$y + 2z = 0, \qquad x + y + z = 1.$$

Hence $y = -2t$ and $x = 1 + t$.

Final result. For $k \neq 3$, there is no solution. For $k = 3$,

$$\mathbf{x} = \begin{bmatrix} 1+t \\ -2t \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

For $k = 3$, substitution gives

$$\begin{aligned} (1+t) - 2t + t &= 1, \\ 2(1+t) + 3(-2t) + 4t &= 2, \\ 3(1+t) + 4(-2t) + 5t &= 3, \end{aligned}$$

so the parameterisation satisfies all three equations.

Marks (8). No-solution classification and rank justification, 2; multiple-solution classification and free-variable justification, 2; correct parametric solution, 2; verification, 2.

Common mistakes. Calling the $k = 3$ case unique; failing to compare rank with the number of unknowns; omitting the parameter domain or verification.

`W02-Q05-PA` **(M4)**

Method and conditions. A single counterexample is enough to disprove a universal claim. Choose

$$\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{Q} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Each matrix is non-zero and has exactly one pivot, so

$$\operatorname{rank}(\mathbf{P}) = \operatorname{rank}(\mathbf{Q}) = 1.$$

Their product is

$$\mathbf{PQ} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

which has rank zero.

In the reverse order,

$$\mathbf{QP} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

which is non-zero and has rank one.

Final result and interpretation. Here

$$\operatorname{rank}(\mathbf{PQ}) = 0 \neq 1 = \min\{\operatorname{rank}(\mathbf{P}), \operatorname{rank}(\mathbf{Q})\}.$$

The example therefore disproves the claim.

It also shows why order matters: $\mathbf{P}$ kills the second-coordinate output of $\mathbf{Q}$, whereas applying $\mathbf{P}$ first leaves a first-coordinate input that $\mathbf{Q}$ sends to a non-zero vector.

Marks (10). Valid matrices, 2; correct two products, 2; correct four ranks, 3; explicit disproof, 1; explanation of the order effect, 2.

Common mistakes. Choosing a zero factor, which does not have rank one; constructing matrices whose forward product still has rank one; overlooking the reverse-product condition; giving an example without calculating all four ranks.

`W02-Q06-PA` **(S2)**

Method and conditions. Apply the multiplication rule on the two branches. Since $\mathbb{P}(E^c) = 2/5$, the four cells are

$$\mathbb{P}(E \cap M) = \mathbb{P}(M \mid E)\mathbb{P}(E) = \frac{2}{3}\frac{3}{5} = \frac{2}{5},$$

$$\mathbb{P}(E \cap M^c) = \frac{1}{3}\frac{3}{5} = \frac{1}{5},$$

$$\mathbb{P}(E^c \cap M) = \mathbb{P}(M \mid E^c)\mathbb{P}(E^c) = \frac{1}{4}\frac{2}{5} = \frac{1}{10},$$

and

$$\mathbb{P}(E^c \cap M^c) = \frac{3}{4}\frac{2}{5} = \frac{3}{10}.$$

Final result. The complete table is

	M	M^c	Total
E	$2/5$	$1/5$	$3/5$
E^c	$1/10$	$3/10$	$2/5$
Total	$1/2$	$1/2$	1

The four joint probabilities sum to one.

Marks (5). Correct four joint probabilities, 3; correct marginal totals, 1; sum-to-one check, 1.

Common mistakes. Reversing the conditional probabilities; using $\mathbb{P}(E)$ on both branches; complementing a marginal rather than the conditional probability; inconsistent row or column totals.

`W02-Q06-PB` **(S2)**

Method and conditions. From the table,

$$\mathbb{P}(M) = \mathbb{P}(E \cap M) + \mathbb{P}(E^c \cap M) = \frac{2}{5} + \frac{1}{10} = \frac{1}{2}.$$

Then

$$\mathbb{P}(E \mid M) = \frac{\mathbb{P}(E \cap M)}{\mathbb{P}(M)} = \frac{2/5}{1/2} = \frac{4}{5}.$$

Similarly,

$$\mathbb{P}(E \mid M^c) = \frac{\mathbb{P}(E \cap M^c)}{\mathbb{P}(M^c)} = \frac{1/5}{1/2} = \frac{2}{5}.$$

Final result and interpretation. The conditional proportions are 4/5 among refresher completers and 2/5 among non-completers. Neither equals the marginal $\mathbb{P}(E) = 3/5$, so E and M are not independent. In this cohort, completing the refresher is positively associated with having an economics background; this probability comparison alone does not establish causation.

Marks (6). Correct $\mathbb{P}(E \mid M)$, 2; correct $\mathbb{P}(E \mid M^c)$, 2; independence conclusion, 1; valid association interpretation, 1.

Common mistakes. Dividing by $\mathbb{P}(E)$; reporting $\mathbb{P}(M \mid E)$; using the same denominator for both conditionals; inferring causation from association.

`W02-Q07-PA` **(S2)**

Method and conditions. Because R_1, R_2, R_3 form a partition, the law of total probability applies. The six joint cells and their totals are

	C	C^c	Total
R_1	3/8	1/8	1/2
R_2	1/6	1/6	1/3
R_3	1/24	1/8	1/6
Total	7/12	5/12	1

For example, $\mathbb{P}(R_1 \cap C) = (1/2)(3/4) = 3/8$, and the other cells follow from the same multiplication rule. Adding the C column gives

$$\begin{aligned}\mathbb{P}(C) &= \sum_{i=1}^{3} \mathbb{P}(C \mid R_i)\mathbb{P}(R_i) \\ &= \frac{3}{4}\frac{1}{2} + \frac{1}{2}\frac{1}{3} + \frac{1}{4}\frac{1}{6} \\ &= \frac{3}{8} + \frac{1}{6} + \frac{1}{24} = \frac{7}{12}.\end{aligned}$$

Final result. $\mathbb{P}(C) = 7/12$.

Marks (5). Correct six joint cells, 3; correct marginal totals, 1; correct total-probability result, 1.

Common mistakes. Adding conditional probabilities without weights; omitting one session; using a conditional probability as a joint table entry; failing to reconcile the marginal totals.

`W02-Q07-PB` **(S2)**

Method and conditions. Bayes' rule reverses the conditioning. Dividing each entry in the C column by $\mathbb{P}(C) = 7/12$ gives

$$\begin{aligned}\mathbb{P}(R_1 \mid C) &= \frac{3/8}{7/12} = \frac{9}{14}, \\ \mathbb{P}(R_2 \mid C) &= \frac{1/6}{7/12} = \frac{2}{7} = \frac{4}{14}, \\ \mathbb{P}(R_3 \mid C) &= \frac{1/24}{7/12} = \frac{1}{14}.\end{aligned}$$

The three posterior probabilities sum to $(9 + 4 + 1)/14 = 1$.

Final result and interpretation. By Bayes' rule, the posterior distribution is $(9/14, 2/7, 1/14)$ over (R_1, R_2, R_3). Session R_1 is the most likely session for a student with a correct worksheet.

Marks (6). Rule named and correctly set up, 1; three correct posterior probabilities, 3; sum check, 1; correct comparison, 1.

Common mistakes. Reporting the supplied conditional probabilities; omitting prior weights; using a different denominator for each posterior; failing to check that the posteriors sum to one.

`W02-Q08-PA` **(S2)**

Method and conditions. The addition rule gives

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) \\ &= \frac{1}{2} + \frac{2}{5} - \frac{7}{10} = \frac{1}{5}.\end{aligned}$$

Also,

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{2}\frac{2}{5} = \frac{1}{5}.$$

Because $\mathbb{P}(B) = 2/5 > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/5}{2/5} = \frac{1}{2} = \mathbb{P}(A).$$

Final result and interpretation. Since $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ and $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, the events are independent under two equivalent tests. They are not disjoint because $\mathbb{P}(A \cap B) = 1/5 > 0$.

Marks (6). Correct intersection, 2; correct conditional probability, 1; two valid independence equalities and conclusion, 2; correct disjointness conclusion, 1.

Common mistakes. Treating the union as the intersection; declaring independent events disjoint; comparing $\mathbb{P}(A \cup B)$ with the product.

`W02-Q09-PA` **(S2)**

Method and conditions. The events D and D^c form a partition, and $\mathbb{P}(D^c) = 19/20$. Total probability gives

$$\begin{aligned}\mathbb{P}(S) &= \mathbb{P}(S \mid D)\mathbb{P}(D) + \mathbb{P}(S \mid D^c)\mathbb{P}(D^c) \\ &= \frac{9}{10}\frac{1}{20} + \frac{1}{10}\frac{19}{20} \\ &= \frac{28}{200}.\end{aligned}$$

Bayes' rule then gives

$$\mathbb{P}(D \mid S) = \frac{(9/10)(1/20)}{28/200} = \frac{9}{28}.$$

For a negative signal, $\mathbb{P}(S^c \mid D) = 1/10$ and $\mathbb{P}(S^c \mid D^c) = 9/10$. Hence

$$\mathbb{P}(S^c) = \frac{1}{10}\frac{1}{20} + \frac{9}{10}\frac{19}{20} = \frac{172}{200} = \frac{43}{50},$$

and

$$\mathbb{P}(D \mid S^c) = \frac{(1/10)(1/20)}{172/200} = \frac{1}{172}.$$

Final result and interpretation. The posterior probability is 9/28 after a positive signal and 1/172 after a negative signal. The positive signal raises the probability of D substantially relative to the prior, while the negative signal lowers it. The value 9/10 is $\mathbb{P}(S \mid D)$, not $\mathbb{P}(D \mid S)$. Because D^c has probability 19/20, false positives from that large group contribute to all positive signals.

Marks (6). Correct positive-signal denominator and posterior, 2; correct negative-signal denominator and posterior, 2; comparison and conditioning-direction explanation, 2.

Common mistakes. Reversing the conditional without Bayes' rule; omitting the false-positive route; failing to complement both signal probabilities for S^c; using 1/20 instead of 19/20 on the complement branch.

`W02-Q10-PA` **(M2 retrieval)**

Method and conditions. Both logarithm arguments must be positive, so $x > 3$. On this domain,

$$\log x + \log(x-3) = \log(x(x-3)).$$

Because log is one-to-one,

$$x(x-3) = 4,$$

so

$$x^2 - 3x - 4 = (x-4)(x+1) = 0.$$

The algebraic roots are 4 and -1, but -1 violates $x > 3$.

Final result. $x = 4$.

Marks (3). Domain restriction, 1; valid logarithm combination and quadratic solution, 1; rejection and final answer, 1.

Common mistakes. Writing $\log x + \log(x-3) = \log(2x-3)$; keeping the inadmissible root; omitting the domain check.

`W02-Q10-PB` **(M3 retrieval)**

Method and conditions. For $f(x) = \log x$,

$$f(1) = 0, \qquad f'(x) = \frac{1}{x}, \qquad f''(x) = -\frac{1}{x^2}.$$

Therefore the second-order Taylor approximation about 1 is

$$T_2(x) = (x-1) - \frac{1}{2}(x-1)^2.$$

At $x = 0.9$ and $x = 1.1$, respectively,

$$\log(0.9) \approx T_2(0.9) = -0.1 - \frac{1}{2}(-0.1)^2 = -0.105,$$
$$\log(1.1) \approx T_2(1.1) = 0.1 - \frac{1}{2}(0.1)^2 = 0.095.$$

Final result and interpretation. $T_2(x) = (x-1) - \frac{1}{2}(x-1)^2$, with approximations -0.105 and 0.095. The quadratic correction is $-\frac{1}{2}(x-1)^2$, which is negative for both non-zero displacements because the displacement is squared and $f''(1) < 0$.

Marks (3). Correct polynomial, 1; both correct approximations, 1; correct sign explanation, 1.

Common mistakes. Omitting the factor $1/2$; using x rather than $x-1$; changing the sign of the quadratic term at $x = 0.9$; giving an equality instead of an approximation.

`W02-Q10-PC (M1; S1 retrieval)`

Method and conditions. The event that at least one of Alice and Benoit is selected is $A \cup B$. By De Morgan's law, its complement is

$$(A \cup B)^c = A^c \cap B^c,$$

the event that neither is selected. Because order is irrelevant, there are

$$\binom{5}{2} = 10$$

equally likely pairs. The complement of selecting at least one of Alice and Benoit is selecting both representatives from the other three students. There are

$$\binom{3}{2} = 3$$

such pairs. Hence the favourable count is $10 - 3 = 7$.

Thus

$$\mathbb{P}(A \cup B) = 1 - \frac{\binom{3}{2}}{\binom{5}{2}} = 1 - \frac{3}{10} = \frac{7}{10}.$$

For a direct count without double-counting, partition the favourable pairs into disjoint cases. Exactly one of Alice and Benoit is selected in $2\binom{3}{1} = 6$ ways, and both are selected in one way. This gives $6 + 1 = 7$ favourable pairs and the same probability.

Final result.

$$\mathbb{P}(A \cup B) = \frac{7}{10}.$$

Marks (2). Correct event and De Morgan complement notation, 1; valid complementary and direct counts with probability $7/10$, 1.

Common mistakes. Confusing union with intersection; writing the De Morgan complement as $A^c \cup B^c$; treating ordered pairs as distinct; double-counting the pair containing both named students.

Self-test solutions and error check

Solutions and error check

Solution to `W02-ST01-PA (M4)` Method. Count pivots in the coefficient block and in the augmented matrix, then compare the common rank with the number of unknowns.

There are two pivots, and the last row is zero on both sides. Therefore

$$\operatorname{rank}(\mathbf{A}) = \operatorname{rank}([\mathbf{A} \mid \mathbf{b}]) = 2 < 3.$$

The system is consistent with multiple solutions. Let $z = t$, where $t \in \mathbb{R}$. The second row gives $y = 1 + t$, and the first gives $x = 4 - 2t$. Hence

$$\mathbf{x} = \begin{bmatrix} 4 - 2t \\ 1 + t \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Marks (5). Two correct ranks, 1; multiple-solution classification with rank justification, 1; valid free parameter, 1; two correct pivot variables, 2.

Common mistakes. Calling the system unique because it is consistent; forgetting that there are three unknowns; omitting $t \in \mathbb{R}$.

Solution to `W02-ST02-PA (M4)` Method. Check invertibility before using the inverse or solving by elimination.

$$\det(\mathbf{D}) = 1(5) - 2(3) = -1 \neq 0,$$

so $\mathbf{D}$ is invertible and the solution is unique. The 2×2 inverse formula gives

$$\mathbf{D}^{-1} = \frac{1}{-1} \begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \mathbf{x} &= \mathbf{D}^{-1}\mathbf{b} \\ &= \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 13 \end{bmatrix} \\ &= \begin{bmatrix} -25 + 26 \\ 15 - 13 \end{bmatrix}. \end{aligned}$$

Thus

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Verification yields

$$\mathbf{D}\mathbf{x} = \begin{bmatrix} 1 + 4 \\ 3 + 10 \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \end{bmatrix} = \mathbf{b}.$$

Marks (5). Correct determinant, 1; invertibility and uniqueness, 1; correct inverse, 1; correct inverse–vector multiplication and solution, 1; verification, 1.

Common mistakes. Using $ad+bc$; not checking that the determinant is non-zero; swapping the solution entries; omitting verification.

Solution to `W02-ST03-PA (S2)` Method. The conditioning event has positive probability, so apply the definition and then compare with the marginal probability.

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/5}{2/5} = \frac{1}{2} = \mathbb{P}(A).$$

Equivalently,

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{2}\frac{2}{5} = \frac{1}{5} = \mathbb{P}(A \cap B).$$

Thus A and B are independent. They are not disjoint because $\mathbb{P}(A \cap B) = 1/5 > 0$.

Marks (5). Correct conditional probability, 2; valid independence test and conclusion, 2; correct disjointness conclusion, 1.

Common mistakes. Dividing by $\mathbb{P}(A)$; treating independence as disjointness; stating a conclusion without a defining equality.

Solution to `W02-ST04-PA (S2)` Method. Use the partition (G, G^c) in total probability, then reverse the conditioning with Bayes' rule.

Since $\mathbb{P}(G^c) = 3/4$,

$$\mathbb{P}(S) = \frac{3}{4}\frac{1}{4} + \frac{1}{4}\frac{3}{4} = \frac{3}{8}.$$

Bayes' rule gives

$$\mathbb{P}(G \mid S) = \frac{\mathbb{P}(S \mid G)\mathbb{P}(G)}{\mathbb{P}(S)} = \frac{(3/4)(1/4)}{3/8} = \frac{1}{2}.$$

The value $\mathbb{P}(S \mid G) = 3/4$ conditions on G, whereas the posterior $\mathbb{P}(G \mid S)$ conditions on S. Reversing the order requires both the prior and the total-probability denominator.

Marks (5). Correct complement branch and total probability, 2; correct Bayes setup, 1; correct posterior, 1; conditioning-direction explanation, 1.

Common mistakes. Omitting the G^c route; using $1/4$ as the denominator; reporting $3/4$ as the posterior.

Record the first cause of each error under `M4` or `S2`, then reattempt that part from a blank page in two or three days.